Tara Friedrich

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Open to opportunities in the SF Bay Area or remote

Summary

Computational scientist with 10+ years of experience in statistics, machine learning, and multi-omics data analysis, turning experimental data into reliable, decision-ready insights. I built the pipelines and databases myself to get closer to the data, which lets me catch quality issues early and trust my interpretation. I have applied these skills to variant interpretation in rare disease and oncology diagnostics, and to prioritizing antibodies for drug candidates using machine learning and structure-to-function models with NGS data. I'm focused on linking structure, function, and other modalities with preclinical and clinical data to connect variants to their biological and patient impact.

Experience

Senior Bioinformatics Engineer - Metaphore Biotechnologies (Flagship Pioneering) March 2025–Present

- Proposed bispecific antibody candidates by analyzing antibody properties (e.g., sequence similarity, developability) from high-throughput screening data to predict heavy and light chain pairing.
- Evaluated embedding-based vs sequence-similarity approaches for candidate prioritization by building a pipeline to cluster 2B antibody sequences (MMseqs2) and benchmarking against PLM embeddings (ESMC, FlashAb, AbLang), validating which analytical approach best supports candidate prioritization across 3 critical pipeline programs.
- Designed and diagnosed issues in antibody screening and maturation (Deep Mutational Scanning) pipelines, reducing turnaround time and improving reproducibility so scientists could iterate faster across discovery campaigns.
- Built an Athena analytics layer on the LIMS data lake to evaluate assay performance and surface experimental trends across campaigns, enabling reanalysis and quality assessment.
- Built interactive dashboards with an LLM-powered component (Claude API) that lets scientists without coding experience explore UMAP embeddings through natural language, with safeguards against unintended data changes, so they can interpret patterns themselves.
- Delivered quarterly project updates and technical/strategic insights in company-wide presentations.

Senior Bioinformatics Scientist - Natera Feb 2022–August 2024

- Partnered with lab scientists to optimize RNA-seq assay performance for detecting oncogenic fusions in noisy FFPE tumor samples, diagnosing unexpected results and improving assay sensitivity and specificity.
- Integrated tumor DNA and RNA data (VAF, clonality, expression, RNA variant support) with HLA binding predictions to rank and select neoantigen peptides for personalized cancer vaccine candidates.
- Developed an integrated classifier to interpret pathogenic genetic variants by combining population genetics, functional annotations, machine learning-based predictors (e.g., AlphaMissense, REVEL, SpliceAI), and curated clinical metadata.
- Built scalable, cloud-native pipelines that reduced turnaround time for WES and RNA-seq QC analyses by 40%, so lab scientists could diagnose assay problems within a single day.

Bioinformatics Scientist - Fabric Genomics (acquired by GeneDx) Nov 2019–Jan 2022

- Maintained and enhanced the ACE algorithm, a production-grade tool for prioritizing clinically pathogenic variants, by implementing comprehensive regression and unit testing to ensure reliability and compliance in a regulated environment.
- Enhanced variant prediction classifiers to increase precision in identifying likely pathogenic variants and flagging borderline cases for use by CAP/CLIA regulated labs using statistics and machine learning.
- Utilized clinical ontologies (HPO, SNOMED, ICD-9/10) to map patient symptoms to structured phenotypes and prioritize candidate variants for rare disease diagnosis and clinical interpretation.
- Leveraged distributed computing frameworks like Apache Spark to scale data classification workflows and optimize ETL processes, streamlining data analysis pipelines for large-scale datasets.
- Actively participated in code reviews, planning sessions, and best practices discussions, fostering a collaborative and high-quality development environment.

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Skills

Programming Languages: Python, R, SQL

Data Science: Pandas, NumPy, Scikit-learn, DuckDB, Polars, PyTorch, Matplotlib, Streamlit, GGplot2, Bioconductor, RShiny

Statistics and Machine Learning: Hypothesis Testing, Linear and Generalized Linear Modeling, Multivariate Modeling, Clustering, Dimensionality Reduction, Supervised and Unsupervised Classification, Random Forests, HMM, Survival Analysis, Representation Learning

Computational Resources: High Performance Cluster, AWS (S3, EC2, Batch, Athena), Google Cloud Compute, Kubernetes, Docker, Argo, WDL, Nextflow, Prefect, Git, SQL (MySQL, Spark), noSQL (MongoDB, DynamoDB), Neo4j, MCP, LangChain

Education

University of California, San Francisco **Ph.D. Bioinformatics**

Statistics and machine learning to identify enhancers (epigenetics) and genes predictive of tissue developmental phenotypes.

University of California, Los Angeles **B.S. Biochemistry**

Publications

Cordero, G.A., Holloway, A.K., **Friedrich, T.**, Eme, J., Eckalbar, W., Kusumi, K., Janzen, F.J., Hicks, J.W., Conlon, F.L., Bruneau, B.G., and Pollard, K.S. (2025). **The Interplay of Ontogeny and Phylogeny at the Transcriptome Level of the Tetrapod Heart.** *Journal of Molecular and Developmental Evolution*.

Friedrich, T.*, Betty, B.*, Mason, M., VanderMeer, J., Zhao, J., Eckalbar, W., Logan, M., Pollard, K.S., Illing, N., and Ahituv, N. (2016). **Bat accelerated regions identify a bat forelimb specific enhancer in the HoxD locus.** *PLoS Genetics*, 12(3):e1005738. (*Co-first authors).

Eckalbar, W., Schlebusch, S., Mason, M., Gill, Z., Booker, B., Nishizaki, S., Nday, C., Terhune, E., Nevonen, K., Makki, N., **Friedrich, T.**, VanderMeer, J., Pollard, K.S., Carbone, L., Wall, J., Illing, N., Parker, A., and Ahituv, N. (2016). **Genomic characterization of the developing bat wing.** *Nature Genetics*, 48, 528–536.

Myers, S.A., Petted, S., Chatterjee, N., **Friedrich, T.**, Tomoda, K., Krings, G., Thomas, S., Broeker, M., Maynard, J., Thomson, M., Pollard, K., Yamanaka, S., Burlingame, A.L., and Panning, B. (2016). **SOX2 O-GlcNAcylation alters its protein-protein interactions and genomic occupancy to modulate gene expression in pluripotent cells.** *eLife*, 5:e10647.

Kostka, D., **Friedrich, T.**, Holloway, A.K., and Pollard, K.S. (2015). **motifDiverge: a model for assessing the statistical significance of gene regulatory motif divergence between two DNA sequences.** *Statistics and Its Interface*, 8(4), 463–476.

Fogel, B.L., Wexler, E., Wahnich, A., **Friedrich, T.**, Vijayendran, C., Gao, F., Parikshak, N., Konopka, G., and Geschwind, D.H. (2012). **RBFOX1 Regulates Both Splicing and Transcriptional Networks in Human Neuronal Development.** *Human Molecular Genetics*, 21(18), 4171–4186.

Konopka, G., **Friedrich, T.**, Davis-Turak, J., Winden, K., Oldham, M.C., Gao, F., Chen, L., Wang, G., Luo, R., Preuss, T.M., and Geschwind, D.H. (2012). **Human-Specific Transcriptional Networks in the Brain.** *Neuron*, 75, 601–617.

Schein, S., Sands-Kidner, M., and **Friedrich, T.** (2008). **The physical basis for the head-to-tail rule that excludes most fullerene cages from self-assembly.** *Biophysical Journal*, 94, 938–957.

Schein, S. and **Friedrich, T.** (2008). **A geometric constraint, the head-to-tail exclusion rule, may be the basis for the isolated-pentagon rule in fullerenes with more than 60 vertices.** *Proceedings of the National Academy of Sciences*, 105, 19142–19147.

Nagarajan, S., **Friedrich, T.**, Garcia, M., Kambham, N., and Sarwal, M.M. (2005). **Gastrointestinal leukocytoclastic vasculitis: an adverse effect of sirolimus.** *Pediatric Transplantation*, 9, 97–100.